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$W_{new} = [1.24, 0.97]$

- Reduces Privacy Concerns
  1. Adds carefully calibrated random noise that hides exact info about individual samples
  2. Encrypt updates before sending them.

## 'SEMI SUPERVISED LEARNING'

→ A machine learning approach that uses:
↳ Small amount of labelled data
↳ and a large amount of unlabelled data

→ Points close to each other are likely to have the same label
→ Samples in the same cluster likely belong to the same class

- Types of Semi Supervised Learn.
  1. Inductive SSL
     ↳ The model learns a general prediction function that can easily training data & unseen future data

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↳ The unlabelled data helps improve the classifier but the final model is meant for future generaliz.

↳ Email Spam Detection

↳ Train using:

- Few labelled emails
- Many unlabeled "

↳ Then classify future incoming emails

↳ Common Methods:

↳ Pseudo - labeling

② Transductive Semi-Supervised Learning

↳ The goal is ONLY to predict labels for the given unlabeled dataset

↳ Doesnot aim to learn a general prediction func for future unseen data

↳ Common Methods:

↳ Graph-based SSL

↳ Label Propagation

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• SSL METHODS

① Self Training
↳ Also called Pseudo-Labeling
↳ It uses small amount of labeled & large amount of unlabeled
↳ Core Idea:
① Learns from labeled data
② generates pseudo-labels for unlabeled data.
③ retrains using those pseudo-labels.
↳ Only predictions with very high confidence are selected
↳ The newly expanded dataset now contains
↳ Original labeled data
↳ High confidence Pseudo-labels
↳ This process is repeated iteratively
↳ Disadvantages:
① Error propagation
② Model reinforces its own mistake repeatedly
③ Sensitive to Confidence threshold
• Too High → very few samples added
• Too Low → noisy labels added

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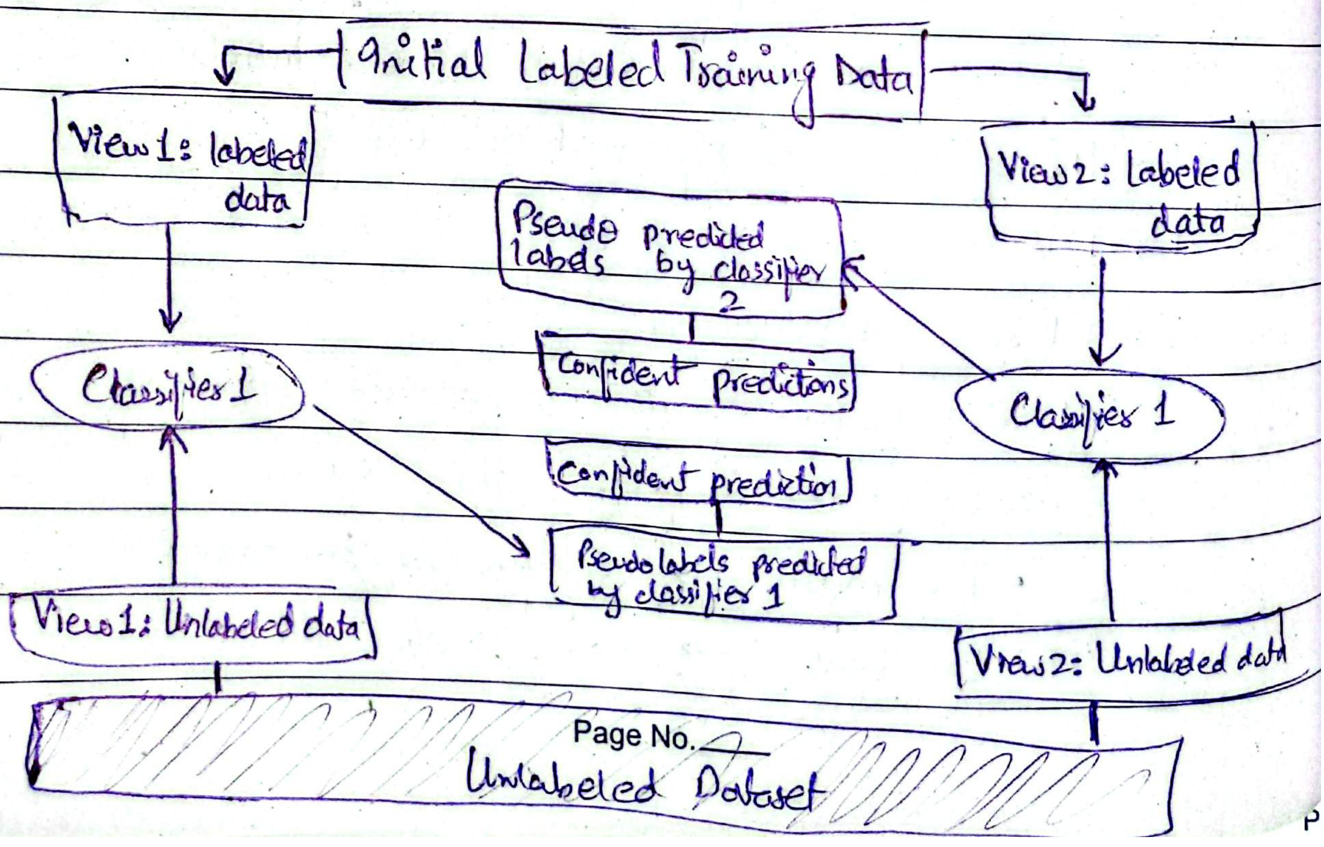
## ② Co-Training

- ↳ Two models are trained on different feature sets and iteratively exchange confident exchange confident predictions to label the unlabeled data
- ↳ The data should be such that it can be split into two featue sets / Views.
- ↳ The two views should be independent given the class labels
- ↳ Disadvantage:
  - ↳ Hard to divide features meaningfully
  - ↳ Wrong Pseudo-labels may spread errors
  - ↳ More computationally expensive



Solutions:
↳ Confidence threshold, keep it 95%
↳ Temperature scaling
↳ Ensemble models

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## ③ Graph-based Label Propagation

↳ Datapoints are represented as nodes in a graph
↳ Similar datapoints are connected
↳ Labels spread from labeled nodes to unlabeled nodes
↳ Components of Graph
- Node ⟶ Data sample
- Edge ⟶ Similarity connection
- Edge Weight → Strength of similarity

↳ Similarity Methods
↳ Euclidean Distance
↳ Cosine similarity
↳ K-nearest neighbour

- Overconfidence Issue in SSL
↳ To certain about its prediction even when those predictions are wrong
↳ Why it Happens?
↳ Only small amount of labeld data available
↳ Class imbalance
↳ Noisy Data

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- **LABEL PROPAGATION EXAMPLE**

↳ We have 3 data points

- $x_1 \rightarrow$ labeled as A
- $x_2 \rightarrow$ " as B
- $x_3 \rightarrow$ unlabeled

**Step#01: Label Matrix $F^{(0)}$**

$$F^{(0)} = \begin{array}{c} \begin{array}{cc} A & B \end{array} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \end{array}$$

**Step2: Similarity Matrix W**

$$W = \begin{bmatrix} 1 & 0.8 & 0.6 \\ 0.8 & 1 & 0.7 \\ 0.6 & 0.7 & 1 \end{bmatrix}$$

**Step3: Degree Matrix D**

Row Sums

$1+0.8+0.6$

$$D = \begin{bmatrix} 2.4 & 0 & 0 \\ 0 & 2.5 & 0 \\ 0 & 0 & 2.3 \end{bmatrix}$$

$0.8+1+0.7$

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Step# 4 : Transition Matrix $T = D^{-1}W$

$$T = \begin{bmatrix} 0.417 & 0.333 & 0.25 \\ 0.320 & 0.4 & 0.28 \\ 0.261 & 0.304 & 0.435 \end{bmatrix}$$

Step#04: First Iteration

$$F^{(1)} = T \cdot F^{(0)}$$

Multiply

~~Results~~ $F^{(1)} = \begin{bmatrix} 0.416 & 0.333 \\ 0.32 & 0.4 \\ 0.260 & 0.304 \end{bmatrix}$

Keeping original labels fixed

$$F^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0.26 & 0.304 \end{bmatrix}$$

2nd iteration

$$F^{(2)} = T \cdot F^{(1)}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0.374 & 0.436 \end{bmatrix}$$

keep then fix

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Class A → 0.374

Class B → 0.436 ✓ Higher probability